INT421—Digital Signal Processing

Linear Time Invariant Systems

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Topics of Last Lecture

- Signal Properties
- Signal Operations
- Special Signals: Impulses and Step Functions
- Complex Sequences

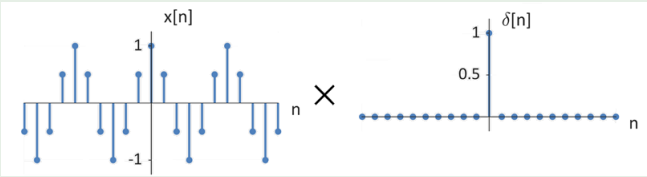
Quiz

Examples

Assume $x[n] = \cos[(n/3)n]$ and let

$$y[n] = x[n]\delta[n]$$

Simplify $y[n]$



Quiz

Solution

Assume $x[n] = \cos[(n/3)n]$ *and let*

$$y[n] = x[n]\delta[n]$$

Simplify $y[n]$

$$y[n] = x[n]\delta[n]$$

$$y[n] = x[0]\delta[n] = \cos[(n/3)n]\delta[n] = \delta[n]$$

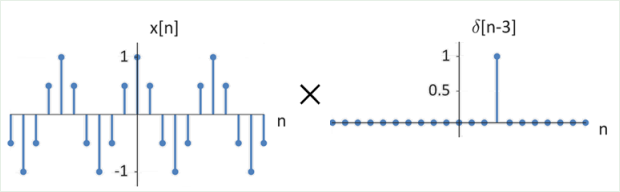
Quiz

Examples

Assume $x[n] = \cos[(n/3)n]$ and let

$$y[n] = x[n]\delta[n-3]$$

Simplify $y[n]$



Quiz

Solution

Assume $x[n] = \cos[(n/3)n]$ *and let*

$$y[n] = x[n]\delta[n]$$

Simplify $y[n]$

$$y[n] = x[n]\delta[n-3]$$

$$y[n] = x[0]\delta[n] = \cos[(n/3)n]\delta[n] = \delta[n-3]$$

Table of Contents

1. Discrete-Time Systems

2. Systems Classifications
   - Memoryless System
   - Causal System
   - Stable System
   - Linear Systems
   - Time-Invariant Systems

3. Summary

Discrete-Time Systems

- Discrete-Time System T is a mathematical operation that maps a given input sequence $x[n]$ into an output sequence $y[n]$

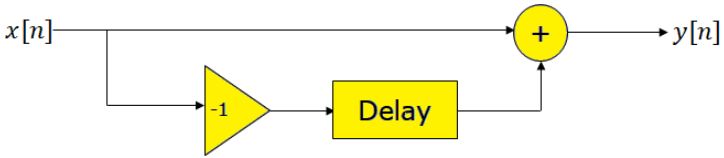
$$y[n] = T(x[n])$$



- Examples of Discrete-Time Systems are:
  1. Moving (Running) Average $y[n] = \frac{1}{4}(x[n] + x[n-1] + x[n-2] + x[n-3])$
  2. Maximum $y[n] = \max\{x[n], x[n-1], x[n-2]\}$
  3. Ideal Delay System $y[n] = x[n - n_0]$

Multiple Representations of Discrete-Time Systems

- Systems can be represented in different ways to more easily address different types of issues
  1. **Verbal description**: 'To reduce the number of bits needed to store a sequence of large numbers that are nearly equal, record the first number, and then record successive differences.'
  2. **Difference equation**:
  $$y[n] = x[n] - x[n-1]$$
  3. **Block diagram**:



  4. We will exploit particular strengths of each of these representations.

Difference Equations

- Difference equations are mathematically precise and compact.
- Example:

$$y[n] = x[n] - x[n-1]$$

- Let $x[n]$ equal the "unit sample" signal $\delta[n]$,

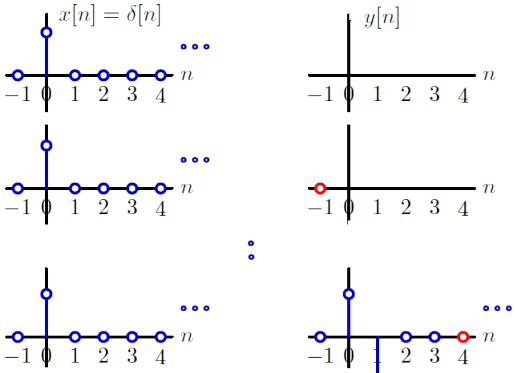
$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

*We use the unit sample as a "primitive" (building-block signal) to construct more complex signals.

Step-By-Step Solution

- Difference equations are convenient for step-by-step analysis.
- Find $y[n]$ given $x[n] = \delta[n]$ and $y[n] = x[n] - x[n-1]$

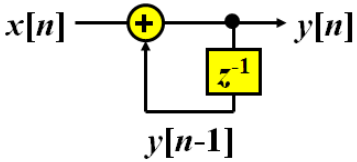
$$y[-1] = x[-1] - x[-2] = 0 - 0 = 0$$
$$y[0] = x[0] - x[-1] = 1 - 0 = 1$$
$$y[1] = x[1] - x[0] = 0 - 1 = -1$$
$$y[2] = x[2] - x[1] = 0 - 0 = 0$$
$$y[3] = x[3] - x[2] = 0 - 0 = 0$$
$$\ldots$$



Accumulator

- Output accumulates all past inputs

$$y[n] = \sum_{l=-\infty}^{n} x[l] = \sum_{l=-\infty}^{n-1} x[l] + x[n] = y[n-1] + x[n]$$



Quiz

Examples

Consider the a **moving average filter** discrete-time system $y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$, find the output sequence, $y[n]$ if the signal $x[n]$ is defined as:

$$x[n] = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \textit{elsewhere} \end{cases}$$

Quiz

Solution

Consider the a **moving average filter** *discrete-time system* $y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$, *find the output sequence,* $y[n]$ *if the signal* $x[n]$ *is defined as:*

$$x[n] = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \textit{elsewhere} \end{cases}$$

$$x[n] = \{\ldots, 3, 2, 1, \underset{\uparrow}{0}, 1, 2, 3, \ldots\}$$

$$y[0] = \frac{1}{3}(x[1] + x[0] + x[-1]) = \frac{1}{3}(1 + 0 + 1) = \frac{2}{3}$$

$$y[n] = \{\ldots, \frac{5}{3}, 2, 1, \underset{\uparrow}{\frac{2}{3}}, 1, 2, \frac{5}{3}, \ldots\}$$

Quiz

Examples

Consider the a **accumulator** discrete-time system $y[n] = (x[n] + x[n-1] + x[n-2] + \dots)$, find the output sequence, $y[n]$ if the signal $x[n]$ is defined as:

$$x[n] = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \textit{elsewhere} \end{cases}$$

Quiz

Solution

Consider the a **accumulator** *discrete-time system* $y[n] = (x[n] + x[n-1] + x[n-2] + \dots)$, *find the output sequence,* $y[n]$ *if the signal* $x[n]$ *is defined as:*

$$x[n] = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \textit{elsewhere} \end{cases}$$

$$x[n] = \{\dots, 3, 2, 1, \underset{\uparrow}{0}, 1, 2, 3, \dots\}$$

$$y[n] = \{\dots, 3, 5, 6, \underset{\uparrow}{6}, 7, 9, 12, 12, \dots\}$$

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Systems Classifications

1. Static (Memoryless) vs. Dynamic (with memory)
2. Causal vs. Non-Causal
3. BIBO Stable vs. BIBO unstable
4. Linear vs. Non-linear
5. Time-invariant vs. Time-variant

Memoryless System

- **Memoryless (Static) System**: A system is memoryless if the output $y[n]$ at every value of n depends only on the input $x[n]$ at the same value of n, i.e., doesn't depend on past or future values of the input.
- **Examples:**
  1. Square: $y[n] = (x[n])^2$
  2. Sign: $y[n] = sign\{x[n]\}$
- **Dynamic Systems or System with Memory**: having either infinite or finite memory
- **Examples:**
  1. Ideal Delay System: $y[n] = (x[n - n_0])$
  2. $y[n] = y[n-1] + x[n]$
  3. $y[n] = x[n] - 4x[n-2]$

Causal System

- **Causality**: A system is causal if it's output is a function of only the current and previous samples, i.e., if, for any n_0, $y[n_0]$ is determined by the values of $x[n]$ for $n < n_0$ only.
- All other systems are non-causal. A subset of non-causal system where the system output, at any time n only depends on future inputs is called anti-causal $y[n] = f[x[n+1], x[n+2], \dots]$.
- **Example:**
  1. Backward Difference:

$$y[n] = x[n] - x[n-1]$$

- **Counter Example**:
  1. Forward Difference

$$y[n] = x[n+1] + x[n]$$

Stable System

- Stability (in the sense of bounded-input bounded-output BIBO).
  - A system is stable if and only if every bounded input produces a bounded output

$$x[n] \leq B_x < \infty \Rightarrow y[n] \leq B_y < \infty$$

- **Example:**
  1. Square:

$$y[n] = (x[n])^2$$

  if input is bounded by $x[n] \leq B_x < \infty$ the output is bounded by $y[n] \leq B_y < \infty$
- **Counter Example**:
  1. Log

$$y[n] = \log_{10}(|x[n]|)$$

  even if input is bounded by $x[n] \leq B_x < \infty$ the output is not bounded by $y[n] = 0 \Rightarrow \log_{10}(|x[n]|) = -\infty$

Linear Systems

- **Linear System**: A system is linear **if and only if**
  - A system is stable if and only if every bounded input produces a bounded output

$$T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_1[n]\} \quad (\textit{additivity})$$

and

$$T\{\alpha x[n]\} = \alpha T\{x[n]\} \quad (\textit{scaling} \quad \textit{and} \quad \textit{homogeneity})$$

- **Combine both: (Superposition Principle)**

$$T\{\alpha_1 x_1[n] + \alpha_2 x_2[n]\} = \alpha_1 T\{x_1[n]\} + \alpha_2 T\{x_1[n]\}$$

Time-Invariant Systems

- Time-Invariant (shift-invariant) Systems
  - A time shift at the input causes corresponding time-shift at output

$$y[n] = T\{x[n]\} \Rightarrow y[n - n_0] = T\{x[n - n_0]\}$$

- **Example:**
  1. Square:

$$y[n] = (x[n])^2$$

  Delay the input the output $y_1[n] = (x[n - n_0])^2$ delay the output gives $y[n - n_0] = (x[n - n_0])^2$
- **Counter Example**:
  1. Compressor System

$$y[n] = x[Mn]$$

  Delay the input the output is $y_1[n] = x[Mn - n_0]$ the delay the output gives $y[n - n_0] = x[M(n - n_0)]$

Quiz

Examples

Are these systems Linear? Time-Invariant? Memoryless? Causal?

1. $y[n] = n^2 x[n]$
2. $y[n] = x[n] - x[n-1]$
3. $y[n] = x[n]x[n-4]$

Quiz

Solution

Are these systems Linear? Time-Invariant? Memoryless? Causal?

1. $y[n] = n^2 x[n]$ *Linear, Time Varying, Memoryless, Causal*
2. $y[n] = x[n] - x[n-1]$ *Linear, Time-Invariant, Has memory, Causal*
3. $y[n] = x[n]x[n-4]$ *Not linear,Time-Invariant, Has memory, Causal*

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Summary

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- System Characteristics
  1. Static (Memoryless) vs. Dynamic (with memory)
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Any Questions?

